GITHUB

NAME: S.GAYATHERI COURSE CODE: CSA1013

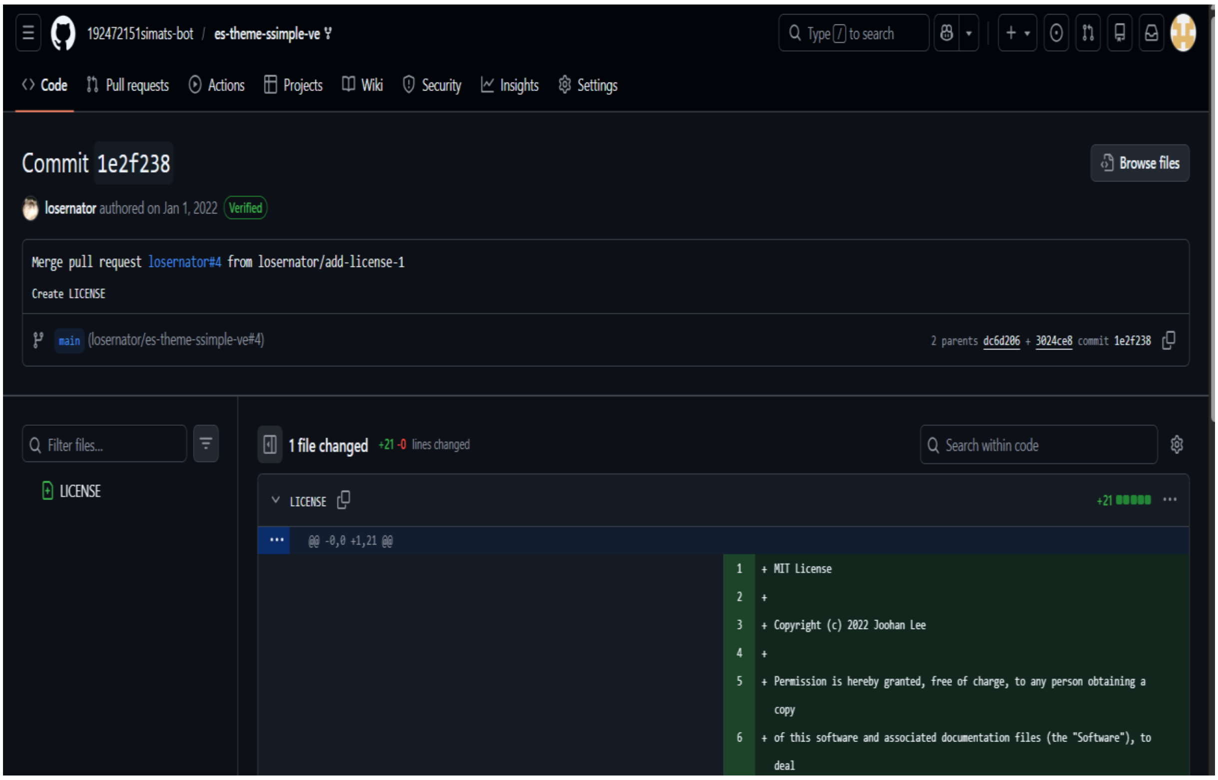
REG NO : 192521306 COURSE NAME: SOFTWARE ENGINEERING

EXP:01- **Collaborative Software Development Using Git and GitHub Through Fork-and-Pull Request Workflow**

Aim

To demonstrate collaborative software development using Git and GitHub through the Fork-and-Pull Request workflow.

Pull Request merged successfully into the main branch.



RESULT

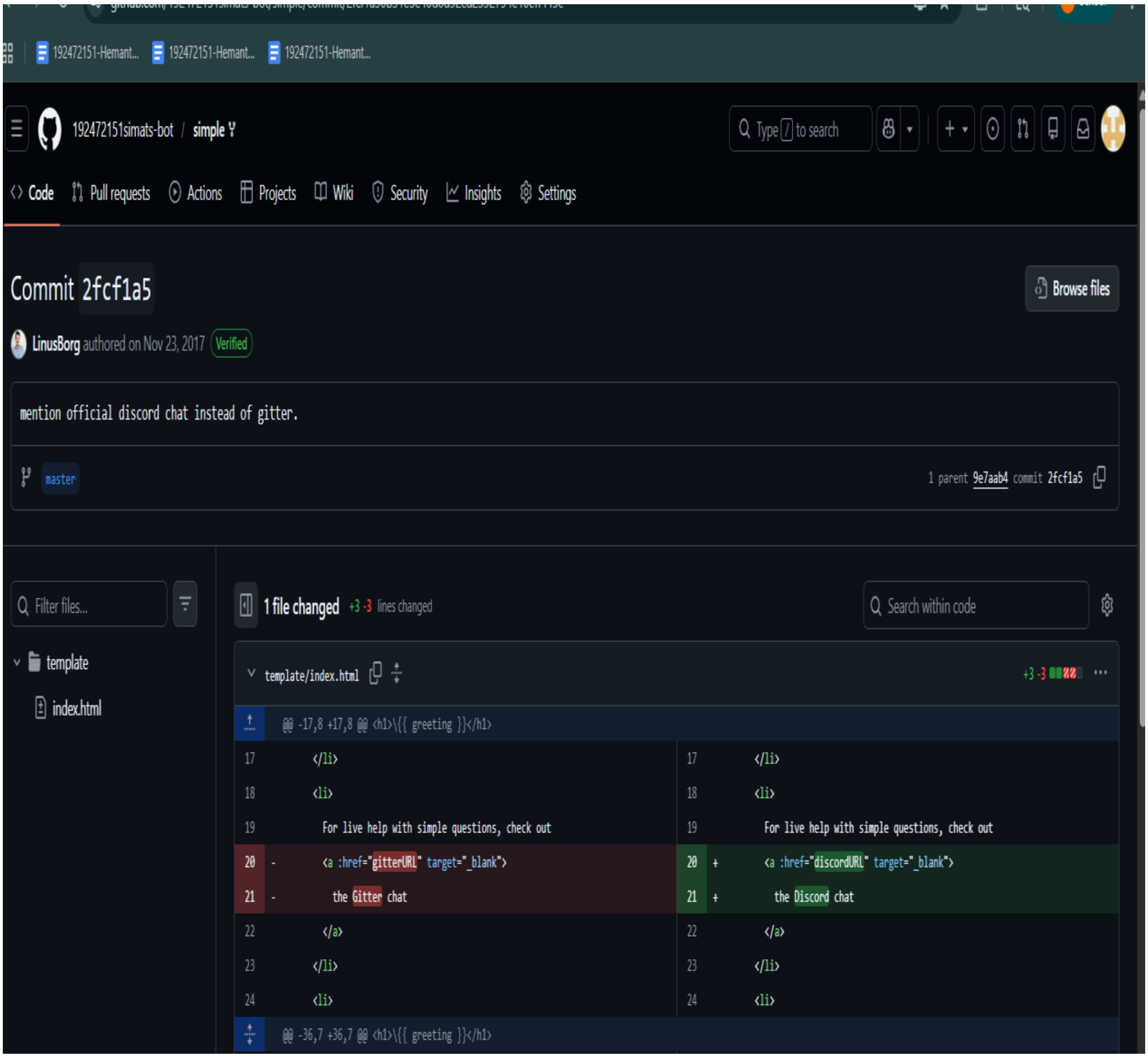
Successfully forked and cloned a GitHub repository, created a feature branch, made and committed changes, pushed the branch to GitHub, created a Pull Request, reviewed the changes, responded to feedback with additional commits, and merged the approved changes into the main branch.

EXP:02-**Demonstrate Git Branching and Merge Conflict Resolution in a Collaborative Project**

Aim

To demonstrate how to create Git branches, collaborate with others, merge changes, and resolve merge conflicts using Git and GitHub.

GitHub commit showing a link change from Gitter to Discord.



Result

Successfully created and worked on a Git branch, merged the latest changes from main, resolved merge conflicts, committed the resolution, pushed the updated branch, and created a Pull Request.

EXP:03- **Create a GitHub Repository and Implement Version Control**

Aim

To create and manage a GitHub repository for a team project using branches, pull requests, code reviews, and conflict resolution.

Git branches created and pushed successfully.

```
C:\Windows\System32\cmd.e
D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 20\TeamProject>
D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 20\TeamProject>git checkout -b backend
Switched to a new branch 'backend'

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 20\TeamProject>git branch
* backend
  frontend
  main

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 20\TeamProject>git checkout frontend
Switched to branch 'frontend'

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 20\TeamProject>git add .

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 20\TeamProject>
D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 20\TeamProject>git commit -m "Added Frontend Module"
On branch frontend
nothing to commit, working tree clean

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 20\TeamProject>
D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 20\TeamProject>git push origin frontend
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'frontend' on GitHub by visiting:
remote:      https://github.com/192524447/TeamProject/pull/new/frontend
remote:
To https://github.com/192524447/TeamProject.git
 * [new branch]      frontend -> frontend

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 20\TeamProject>git commit -m "Added Frontend Module"
On branch frontend
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        frontend.txt

nothing added to commit but untracked files present (use "git add" to track)

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 20\TeamProject>git push origin frontend
Everything up-to-date

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 20\TeamProject>
```

Result

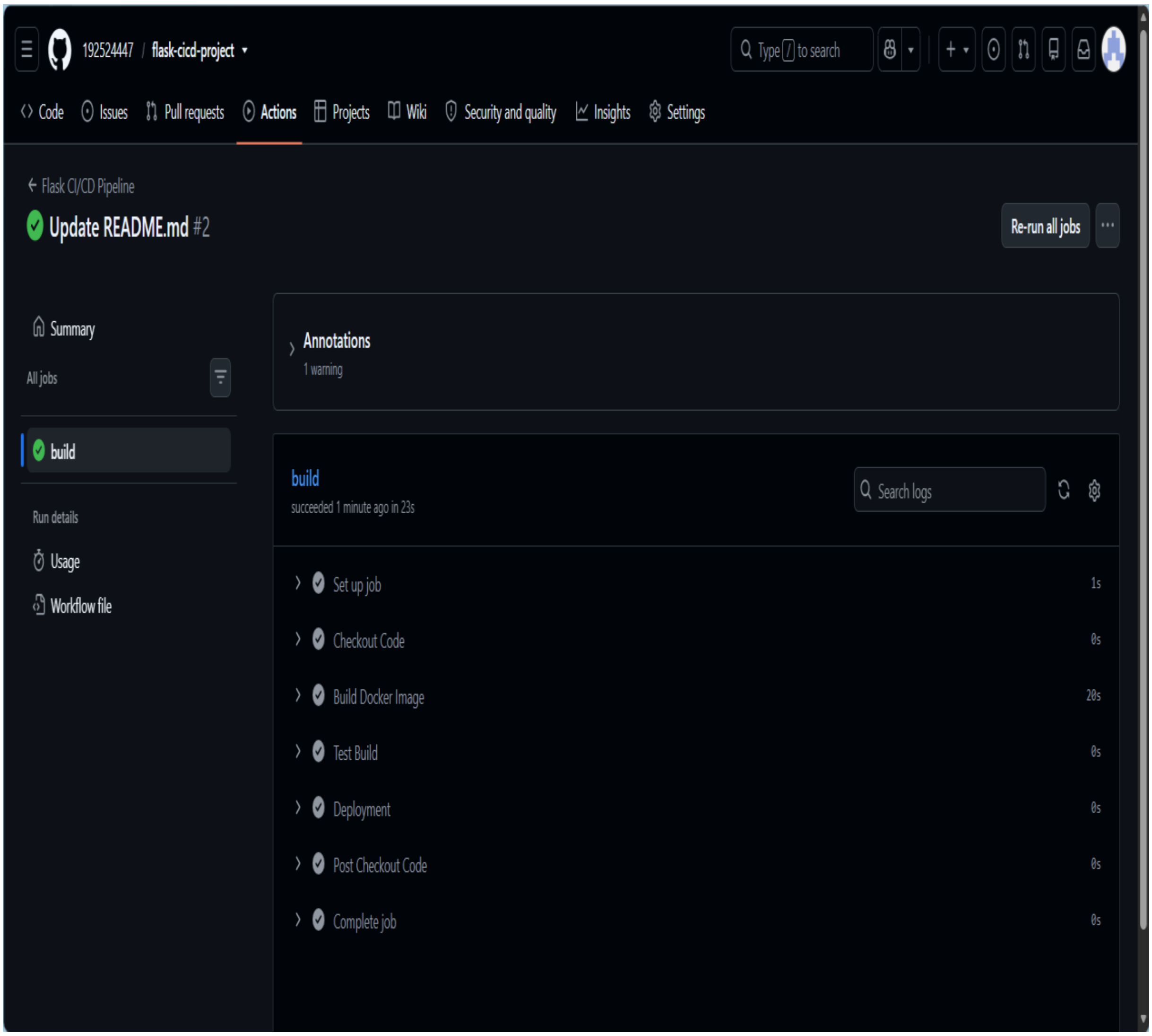
Successfully created a GitHub repository, implemented module-wise branches, merged changes through Pull Requests and code reviews, resolved merge conflicts, and documented the version-control workflow in the README file.

EXP:04- **Create a CI/CD Pipeline Using GitHub Actions**

Aim

To automate the testing, Docker image building, publishing, and deployment of a containerized Flask application using GitHub Actions.

CI/CD pipeline completed successfully.



Result

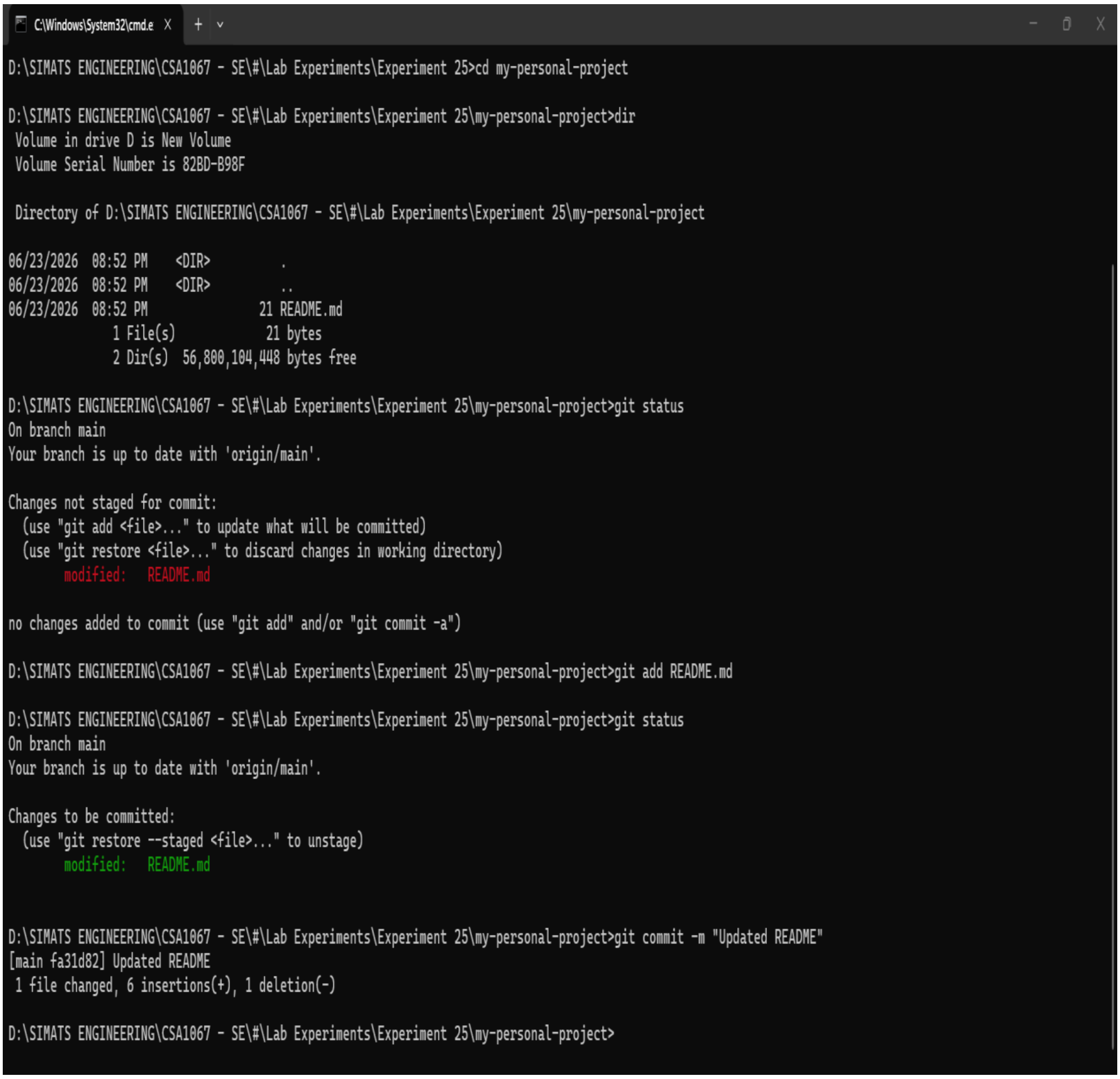
Successfully created a CI/CD pipeline for a containerized Flask application using GitHub Actions, automated testing and Docker image creation, pushed the image to Docker Hub, and deployed the application to a cloud platform.

EXP:05- **Create and Manage a Personal GitHub Repository**

Aim

To create a GitHub repository, clone it locally, modify the README.md file, and push the changes back to GitHub.

README updated and committed successfully



```
C:\Windows\System32\cmd.e

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 25>cd my-personal-project

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 25\my-personal-project>dir
 Volume in drive D is New Volume
 Volume Serial Number is 82BD-B98F

 Directory of D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 25\my-personal-project

06/23/2026  08:52 PM    <DIR>          .
06/23/2026  08:52 PM    <DIR>          ..
06/23/2026  08:52 PM                21 README.md
               1 File(s)             21 bytes
               2 Dir(s)  56,800,104,448 bytes free

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 25\my-personal-project>git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 25\my-personal-project>git add README.md

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 25\my-personal-project>git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md


D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 25\my-personal-project>git commit -m "Updated README"
[main fa31d82] Updated README
 1 file changed, 6 insertions(+), 1 deletion(-)

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 25\my-personal-project>
```

Result

Successfully created a GitHub repository, cloned it locally, updated the README file, and pushed the changes to GitHub.

EXP:06- **Clone a GitHub Repository and Make a Small Change**

Aim

To clone an existing GitHub repository and modify a file locally.

README.md modified successfully.

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 26>git clone https://github.com/192524447/my-personal-project.git
Cloning into 'my-personal-project'...
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 9 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (9/9), done.

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 26>cd my-personal-project

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 26\my-personal-project>dir
 Volume in drive D is New Volume
 Volume Serial Number is 82BD-B98F

 Directory of D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 26\my-personal-project

06/23/2026  09:51 PM    <DIR>          .
06/23/2026  09:51 PM    <DIR>          ..
06/23/2026  09:51 PM                23 README.md
               1 File(s)             23 bytes
               2 Dir(s)  56,799,596,544 bytes free

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 26\my-personal-project>git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 26\my-personal-project>
```

Result

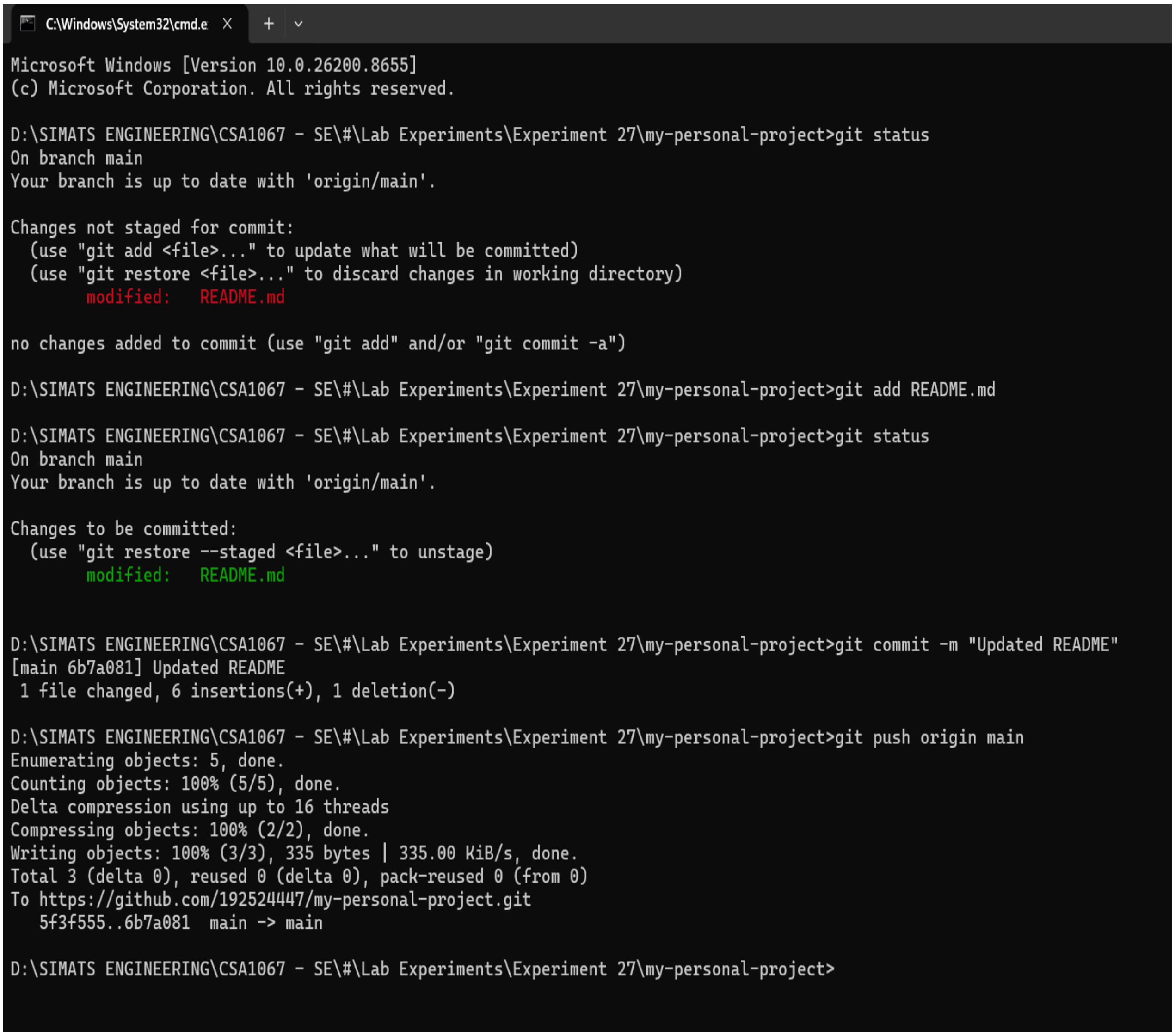
Successfully cloned the GitHub repository and made a small change to a project file.

EXP:07**- Commit and Push Changes to GitHub**

Aim

To commit changes made to a cloned repository locally and push them to GitHub using Git.

Changes committed and pushed to GitHub.



```
C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 27\my-personal-project>git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 27\my-personal-project>git add README.md

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 27\my-personal-project>git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md


D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 27\my-personal-project>git commit -m "Updated README"
[main 6b7a081] Updated README
 1 file changed, 6 insertions(+), 1 deletion(-)

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 27\my-personal-project>git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 335 bytes | 335.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/192524447/my-personal-project.git
   5f3f555..6b7a081  main -> main

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 27\my-personal-project>
```

Result

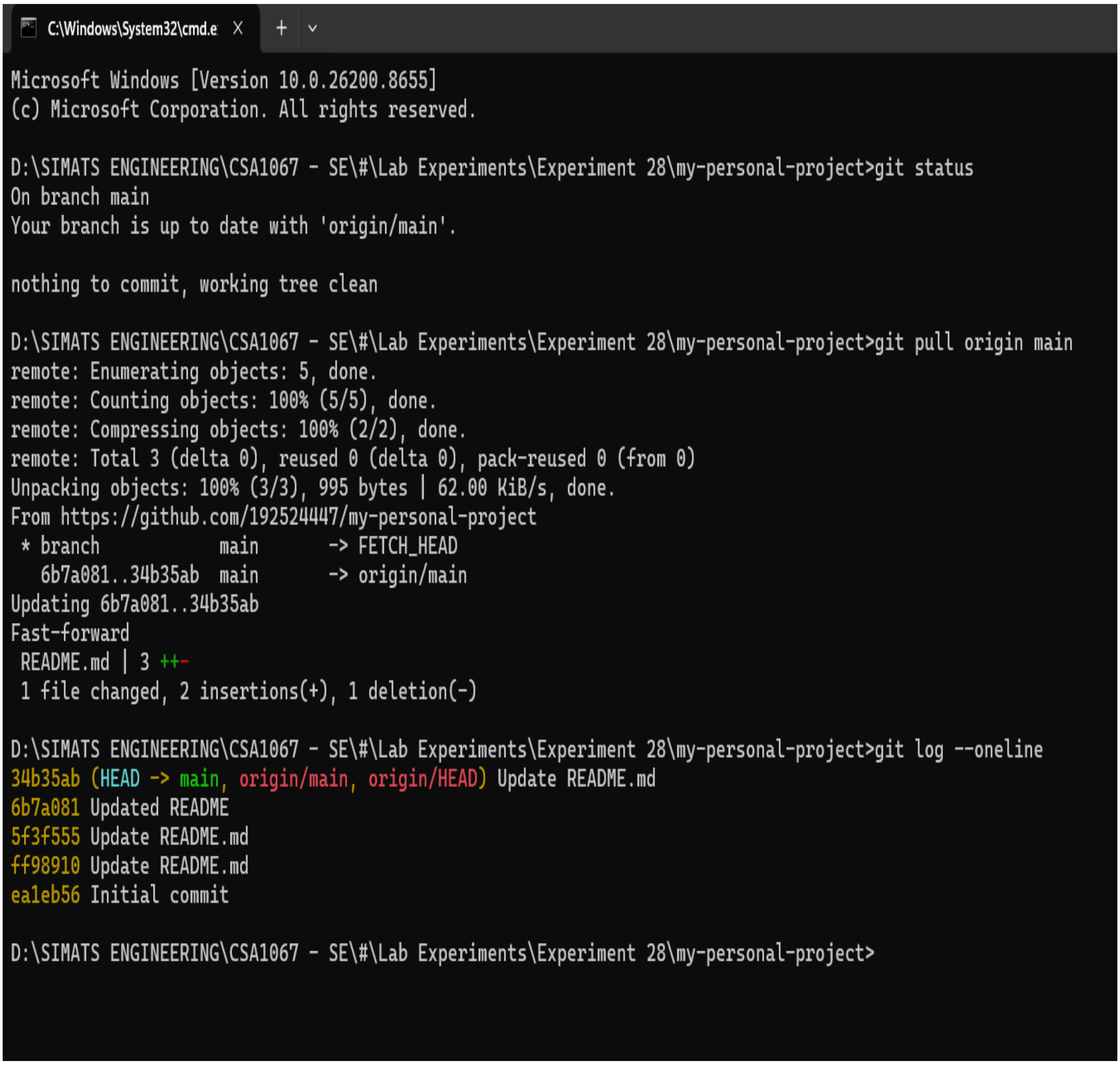
Changes were successfully committed locally and pushed to GitHub.

EXP:08- **Pull Latest Changes from a GitHub Repository**

Aim

To update the local repository with the latest changes made by a collaborator on GitHub**.**

Latest changes pulled successfully.



```
C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 28\my-personal-project>git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 28\my-personal-project>git pull origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 995 bytes | 62.00 KiB/s, done.
From https://github.com/192524447/my-personal-project
 * branch            main       -> FETCH_HEAD
   6b7a081..34b35ab  main       -> origin/main
Updating 6b7a081..34b35ab
Fast-forward
 README.md | 3 ++-
 1 file changed, 2 insertions(+), 1 deletion(-)

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 28\my-personal-project>git log --oneline
34b35ab (HEAD -> main, origin/main, origin/HEAD) Update README.md
6b7a081 Updated README
5f3f555 Update README.md
ff98910 Update README.md
ea1eb56 Initial commit

D:\SIMATS ENGINEERING\CSA1067 - SE\#\Lab Experiments\Experiment 28\my-personal-project>
```

Result

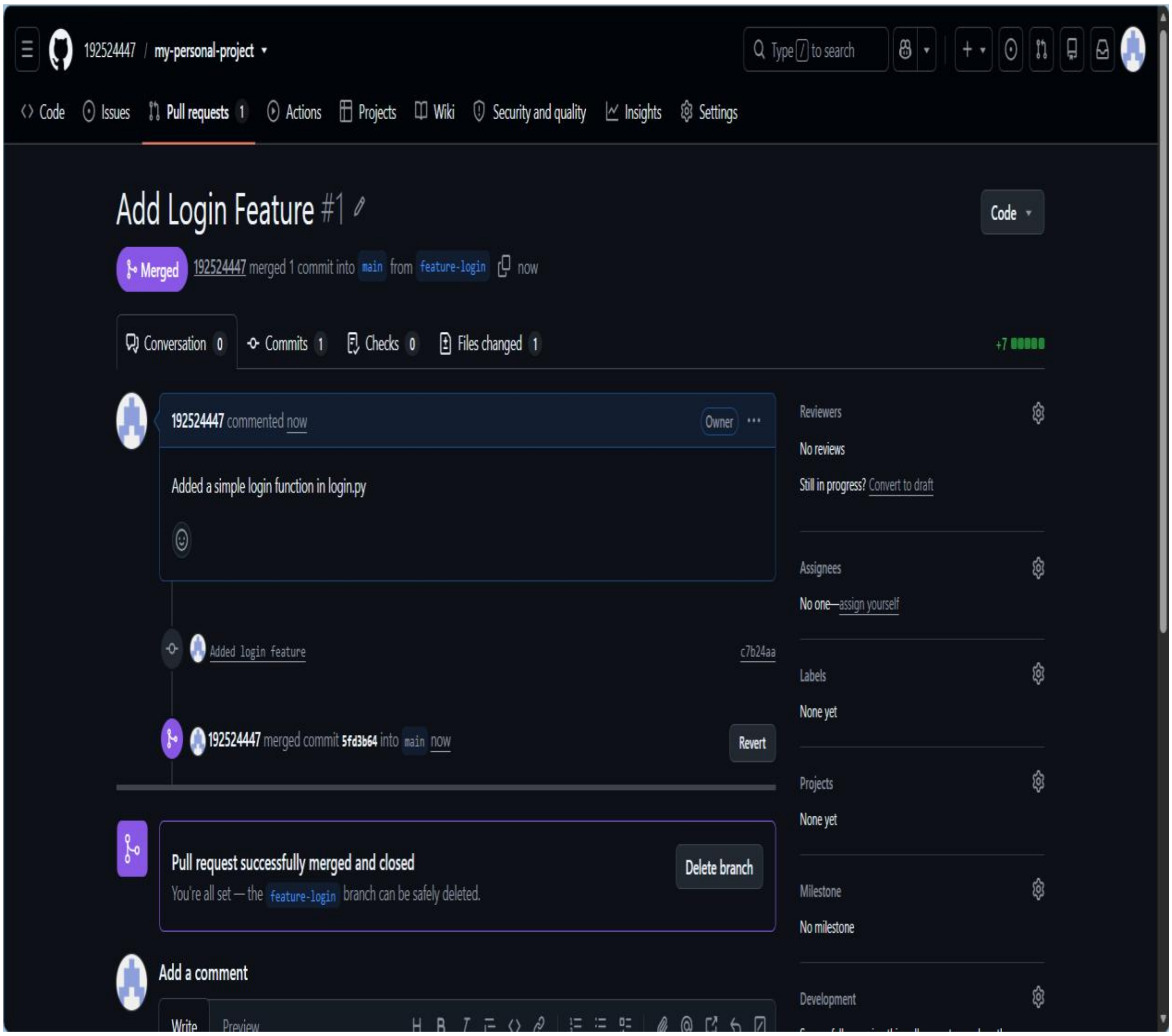
Latest collaborator changes were successfully pulled from GitHub and updated in the local repository.

EXP:09- **Create a Feature Branch and Merge a Pull Request**

Aim

To create a feature-login branch, implement a simple login function, and merge it into the main branch using a Pull Request.

Login feature merged successfully.



Result

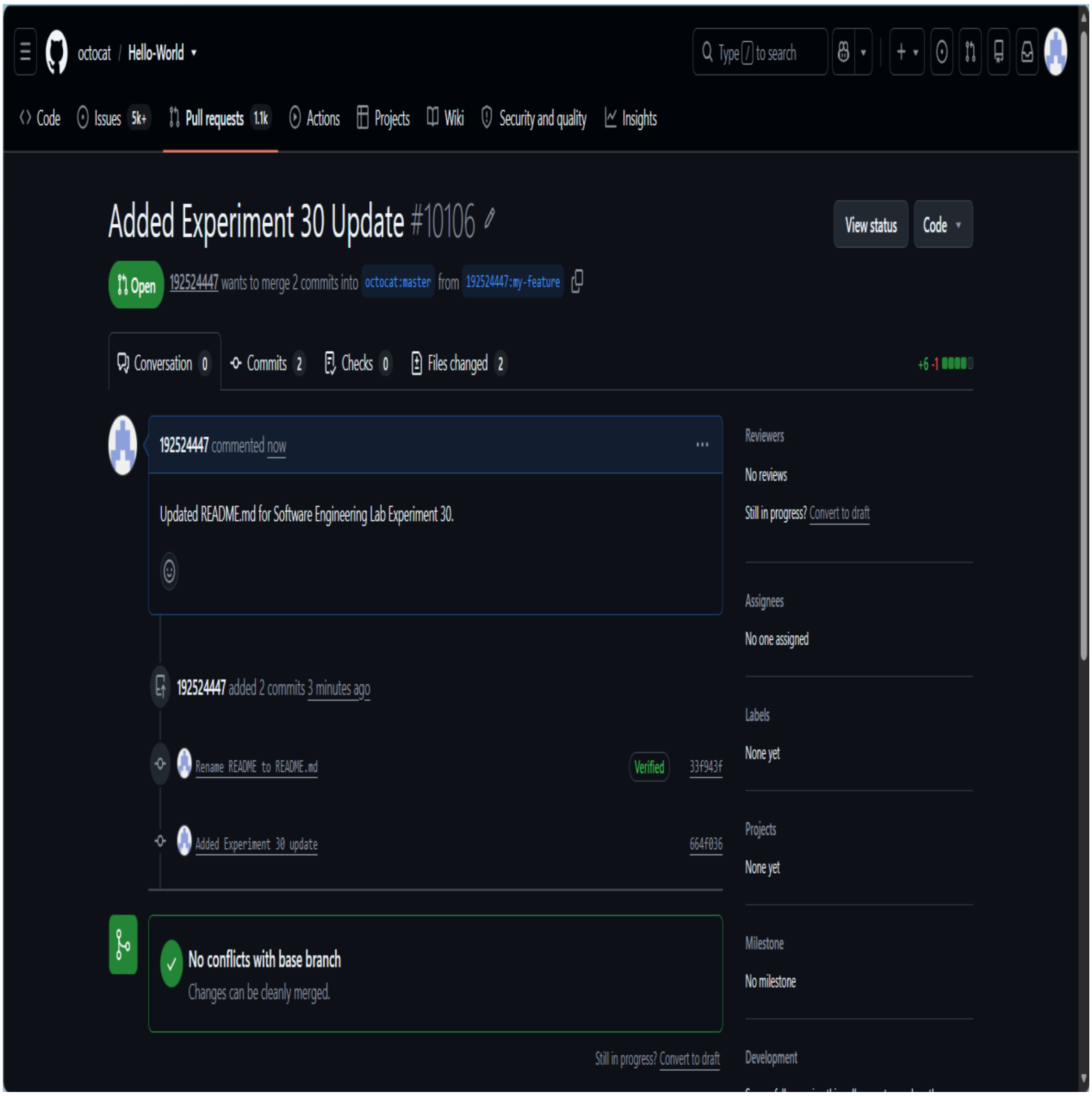
The feature-login branch was created, the login function was added, and the changes were successfully merged into main.

EXP:10- **Fork a Repository and Submit a Pull Request**

Aim

To fork a repository, create a feature branch, implement changes, and submit a Pull Request to the original repository.

Pull request created successfully with no conflicts.



Result

Successfully forked the repository, created a feature branch, implemented changes, pushed them to the fork, and submitted a Pull Request to the original repository.

Google Sheet or Excel

EXP: 11-**MoSCoW Prioritization of Library Management System Requirements**

Aim:
To categorize the Library Management System requirements using the MoSCoW method based on user impact, feasibility, time, budget, and resources.

MoSCoW categorization of library system requirements in Excel.

Req ID	Requireme	MoSCoW (	Justificatic	Priority Reason
R1	Search bo	Must-Hav	Core librai	Essential for users
R2	Online bo	Must-Hav	Improve a	High user demand
R3	Monthly r	should-Ha	Important	Operational efficiency
R4	Email ove	Must-Hav	Prevents b	High impact
R5	QR Code	Could-Hav	Enhances	Erequires extra hardware
R6	Librarian c	Must-Hav	Required f	Core System control
R7	Review an	Could-Hav	Improves	optional feature
R8	Chatbot	Could-Hav	Helps user	Resource intensive
R9	Mobile ap	should-Ha	improves	Medium feasibility
R10	Multi-lang	Won't-Ha	High cost	Low priority initially

Result:
All requirements were successfully classified as Must-Have, Should-Have, Could-Have, or Won't-Have, with suitable justifications.